

DSA 210 TERM PROJECT
FINAL REPORT

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1. Motivation

Urban transportation is an integral part of daily life and travel time is often perceived as unpredictable due to various factors such as traffic conditions, time of day, and route characteristics. Understanding these factors is important not only for improving personal time management but also for gaining insights into broader urban mobility patterns.

This project is motivated by the need to better understand the determinants of daily travel time based on real-life driving experiences. While navigation applications such as Google Maps provide estimated travel durations, these estimates may not always reflect actual conditions. This creates an opportunity to investigate the relationship between estimated and actual travel times and to evaluate the reliability of similar tools.

Additionally, by analyzing how variables like traffic level, departure time, and distance affect trip duration, this project aims to identify key patterns that influence travel efficiency. The use of self-collected data makes the analysis more personalized and realistic, allowing for a deeper exploration of everyday driving behavior.

Ultimately, the motivation behind this study is to apply data science techniques to a familiar real-world problem, transforming daily observations into meaningful insights and predictive understanding of travel time dynamics.

2. Data Source

The dataset used in this project was collected manually through real-life driving experiences, as outlined in the project proposal. The data gathering process was conducted between April 1 and April 14 and includes trips made by the author, as well as by family members and friends. This approach allowed the dataset to reflect diverse driving patterns under varying conditions.

For each trip, multiple variables were recorded to capture different aspects influencing travel time, the first 5 rows are given on Figure 1. These include departure time, arrival time, actual travel duration, distance, number of traffic lights, traffic level (categorized on a scale), and whether the trip occurred during rush hours. Additionally, estimated travel time values were obtained from Google Maps to enable a comparison between predicted and actual trip durations.

date	departure	arrival	actual_duration	destination	num_lights	distance	estimated_duration	traffic_level
1.04.2026	13:32	14:48	01:16	Florya - Tuzla	11	64 km	01:14	Medium
1.04.2026	18:54	19:11	00:17	Tuzla - Şekerpinar	1	6.5 km	00:10	Low
1.04.2026	21:24	22:09	00:45	Tuzla - Yeşilköy	4	61 km	00:53	Low
2.04.2026	12:05	13:13	01:08	Florya - Tuzla	12	64 km	01:09	Medium
2.04.2026	16:46	19:07	02:21	Tuzla - Florya	7	64 km	02:01	High

Figure 1

The dataset consists of approximately 100–150 observations, where each row represents a single trip and each column corresponds to a specific feature. Since the data is self-collected, it provides a realistic and personalized view of urban driving behavior, though it may also reflect certain biases related to the sample group and geographic area.

Overall, this dataset is suitable for exploratory data analysis, hypothesis testing, and the application of basic machine learning models to understand and predict travel time dynamics.

3. Data Analysis

The analysis in this project was conducted in multiple stages, following a structured data science pipeline including data preprocessing, exploratory data analysis (EDA), statistical testing, and machine learning modeling.

3.1. Data Preprocessing

Initially, the dataset was cleaned and transformed into a suitable format for analysis. Time-related variables such as departure and arrival times were converted into datetime format, and travel durations were standardized in minutes, initial and final data formats are given in Figure 2 and Figure 3 respectively.

First 5 rows:

	date	departure	arrival	actual_duration	\
0	2026-03-31 00:00:00	07:49:00	08:30:00	00:41:00	
1	2026-03-31 00:00:00	17:25:00	18:16:00	00:51:00	
2	2026-03-31 00:00:00	18:31:00	18:36:00	00:05:00	
3	2026-03-31 00:00:00	20:17:00	20:21:00	00:04:00	
4	2026-04-01 00:00:00	13:32:00	14:48:00	01:16:00	

	destination	num_lights	distance	estimated_duration	traffic_level
0	Florya - Altunizade	13	29 km	00:44:00	Medium
1	Altunizade - Florya	8	31 km	00:53:00	High
2	Florya - Florya	0	0.6 km	00:02:00	Low
3	Florya - Florya	0	1.1 km	00:03:00	Low
4	Florya - Tuzla	11	64 km	01:14:00	Medium

Figure 2

Additional variables such as delay (difference between estimated and actual duration) and rush hour indicators were also derived.

- Delay: calculated as the difference between actual and estimated duration
- Rush hour indicator: a binary variable representing peak hours

Final Data:

	date	departure	arrival	actual_duration	\
0	2026-03-31 2026-05-05	07:49:00	08:30:00	00:41:00	
1	2026-03-31 2026-05-05	17:25:00	18:16:00	00:51:00	
2	2026-03-31 2026-05-05	18:31:00	18:36:00	00:05:00	
3	2026-03-31 2026-05-05	20:17:00	20:21:00	00:04:00	
4	2026-04-01 2026-05-05	13:32:00	14:48:00	01:16:00	

	destination	num_lights	distance	estimated_duration	traffic_level	\
0	Florya - Altunizade	13	29.0	00:44:00	Medium	
1	Altunizade - Florya	8	31.0	00:53:00	High	
2	Florya - Florya	0	0.6	00:02:00	Low	
3	Florya - Florya	0	1.1	00:03:00	Low	
4	Florya - Tuzla	11	64.0	01:14:00	Medium	

	actual_duration_min	estimated_duration_min	rush_hour	delay_min
0	41.0	44.0	Yes	-3.0
1	51.0	53.0	Yes	-2.0
2	5.0	2.0	Yes	3.0
3	4.0	3.0	No	1.0
4	76.0	74.0	No	2.0

Figure 3

The final dataset consisted of 127 observations and 9 variables, with no missing values, ensuring data completeness and reliability for further analysis.

```
Shape: (127, 9)

Columns:
Index(['date', 'departure', 'arrival', 'actual_duration', 'destination',
       'num_lights', 'distance', 'estimated_duration', 'traffic_level'],
      dtype='object')

Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 127 entries, 0 to 126
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   date                  127 non-null   object
1   departure              127 non-null   object
2   arrival                127 non-null   object
3   actual_duration        127 non-null   object
4   destination            127 non-null   object
5   num_lights             127 non-null   int64
6   distance               127 non-null   object
7   estimated_duration     127 non-null   object
8   traffic_level          127 non-null   object
dtypes: int64(1), object(8)
memory usage: 9.1+ KB
None
```

Figure 4

```
Missing values:
date           0
departure      0
arrival        0
actual_duration 0
destination    0
num_lights     0
distance       0
estimated_duration 0
traffic_level  0
dtype: int64
```

Figure 5

3.2. Exploratory Data Analysis (EDA)

3.2.1. Distribution Analysis Histograms

The distribution of actual travel duration is right-skewed as shown in Figure 6, indicating that most trips are relatively short while a small number of trips take significantly longer. The majority of observations fall between 10 and 60 minutes, with a few extreme values representing longer trips.

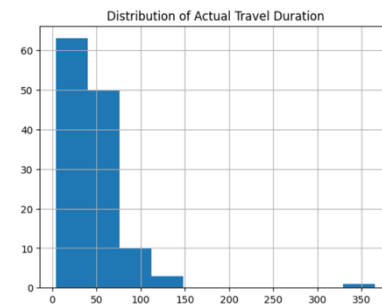


Figure 6

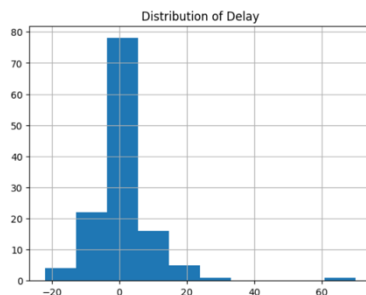


Figure 7

The delay distribution is centered around zero, indicating that estimated travel times are generally accurate. As shown in Figure 7 most values fall within a narrow range around zero, meaning that the difference between estimated and actual durations is typically small.

However, the distribution includes both negative and positive values:

- Negative values indicate that the estimated time was higher than the actual time (overestimation)
- Positive values indicate underestimation

The presence of a few extreme values suggests that, in some cases, significant deviations occur, but these are relatively rare.

3.2.2. Correlation Analysis

A strong positive correlation (≈ 0.89) was observed between distance and actual travel duration, indicating that travel time increases proportionally with distance. This confirms that distance is the dominant factor influencing trip duration. (Figure 8)

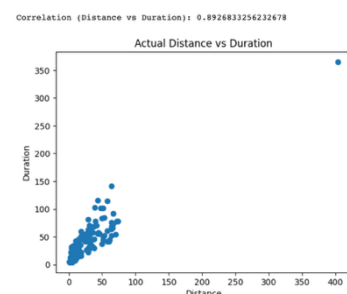
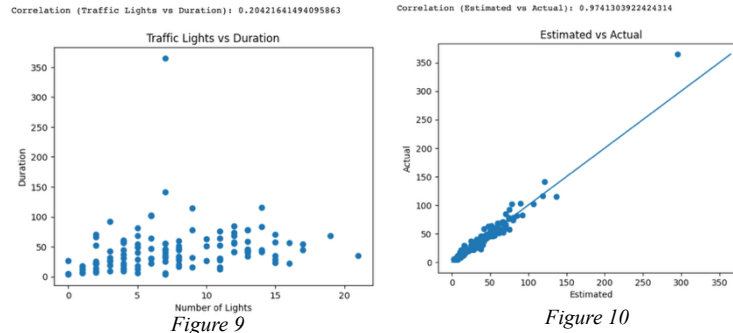


Figure 8

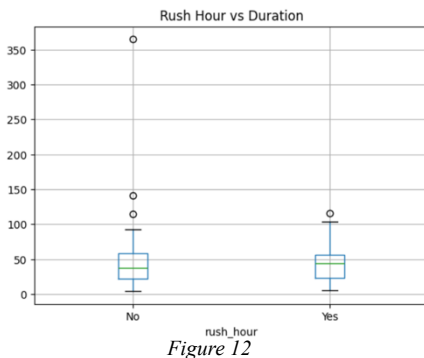
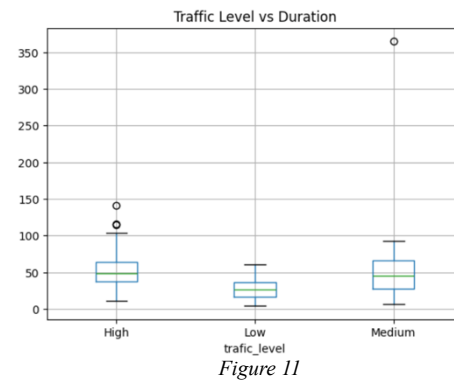
The relationship between number of traffic lights and travel duration is relatively weak (≈ 0.20), suggesting that traffic lights alone are not a strong determinant of travel time. (Figure 9)

Additionally, a very strong correlation (≈ 0.97) exists between estimated and actual durations, demonstrating that navigation tools such as Google Maps provide highly reliable predictions. (Figure 10)



3.2.3. Comparative Analysis

The boxplot shows a clear relationship between traffic level and travel duration. Trips under low traffic conditions have noticeably lower median durations and a tighter distribution, indicating more consistent and shorter travel times. In contrast, medium and high traffic levels exhibit higher median durations and greater variability. This suggests that as traffic intensity increases, not only does travel time increase, but it also becomes less predictable.



The boxplot comparing rush hour and non-rush hour trips does not show a clear separation between the two groups. The medians are relatively close, and the spread of the data is similar for both categories. The distributions largely overlap, suggesting that rush hour alone is not a strong determinant of travel duration.

3.2.4. Traffic Level Based Analysis

The average values further confirm the significant impact of traffic conditions on travel duration. As traffic intensity increases from low to high levels, travel times also increase noticeably. Trips occurring under medium traffic conditions take nearly twice as long as those experienced during low traffic periods. Additionally, high traffic conditions produce the longest travel durations overall, demonstrating how congestion directly affects transportation efficiency and commuting time.

```
Average duration by traffic level:
traffic_level
High      58.034483
Low       26.880952
Medium    51.607143
Name: actual_duration_min, dtype: float64
```

Figure 13

The delay analysis reveals that prediction accuracy changes depending on traffic conditions. Under low traffic conditions, travel durations tend to be slightly overestimated, resulting in negative delay values. In contrast, medium and high traffic conditions show slight underestimation, which leads to positive delay values. Although the direction of prediction errors varies across different traffic levels, the overall differences remain relatively small, indicating that the estimations are generally close to actual travel durations.

Average delay by traffic level:
 traffic_level
 High 1.275862
 Low -0.285714
 Medium 1.803571
 Name: delay_min, dtype: float64

Figure 14

3.3. Hypothesis Testing

To statistically validate the observed patterns, several hypothesis tests were conducted.

3.3.1. Independent T-test: Rush Hour vs Non-Rush Hour

H0: There is no significant difference in actual travel duration between rush hour and non-rush hour trips.
 H1: There is a significant difference in actual travel duration between rush hour and non-rush hour trips.
 Rush hour mean: 43.785714285714285
 Non-rush hour mean: 45.44705882352941
 t-statistic: -0.2632160958002544
 p-value: 0.792841295508658

Figure 15

The p-value was greater than 0.05, leading to a failure to reject the null hypothesis. This indicates that rush hour does not have a statistically significant effect on travel duration in this dataset.

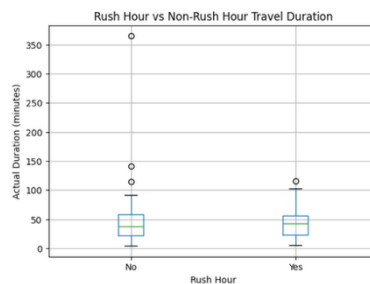


Figure 16

3.3.2. ANOVA Test: Traffic Level

H0: Mean actual travel duration is the same across traffic levels.
 H1: At least one traffic level has a different mean actual travel duration.
 F-statistic: 7.773549942802812
 p-value: 0.0006598632376532952

Figure 17

The p-value was less than 0.05, leading to rejection of the null hypothesis. This shows that traffic level has a statistically significant impact on travel duration.

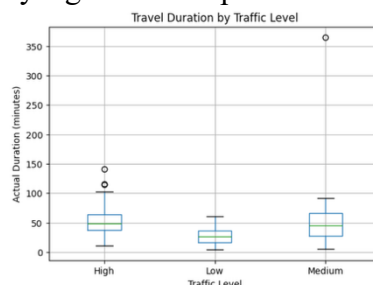


Figure 18

3.3.3. Paired T-test: Estimated vs. Actual Duration

H0: There is no significant difference between actual and estimated travel durations.
H1: There is a significant difference between actual and estimated travel durations.
Actual mean: 44.89763779527559
Estimated mean: 43.90551181102362
t-statistic: 1.1979862916737511
p-value: 0.23317098509478104

Figure 19

The test resulted in a p-value greater than 0.05, indicating no statistically significant difference between estimated and actual travel times. This suggests that Google Maps estimates are generally accurate.

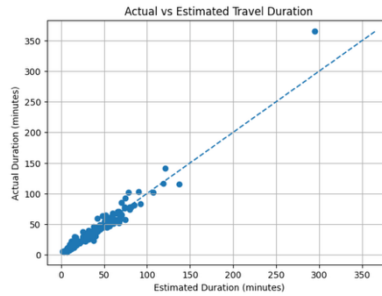


Figure 20

3.3.4. Pearson Correlation Tests: Distance and Number of Traffic Lights vs. Actual Travel Duration

Distance showed a statistically significant strong relationship with travel duration ($p < 0.05$), while the number of traffic lights also showed a statistically significant but weaker relationship.

H0: There is no linear relationship between distance and actual travel duration.
H1: There is a significant linear relationship between distance and actual travel duration.
Correlation coefficient: 0.8926833256232681
p-value: 4.3164259901092526e-45

Figure 21

H0: There is no linear relationship between distance and actual travel duration.
H1: There is a significant linear relationship between distance and actual travel duration.
Correlation coefficient: 0.8926833256232681
p-value: 4.3164259901092526e-45

Figure 22

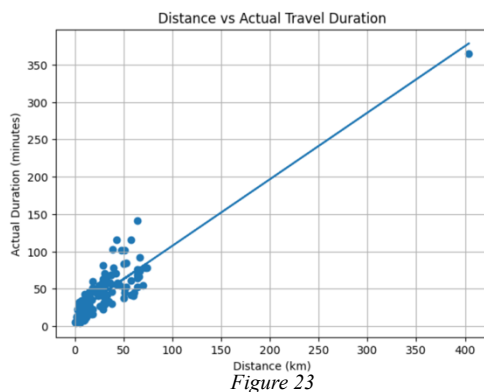


Figure 23

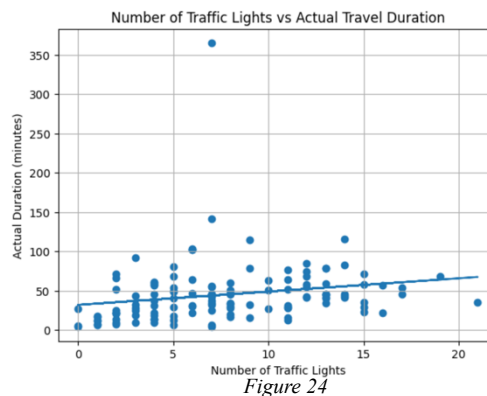


Figure 24

3.3.5. One-sample T-test: Delay

The average delay was not significantly different from zero, further supporting the reliability of estimated travel times.

```
H0: The average delay is equal to 0.  
H1: The average delay is significantly different from 0.  
Average delay: 0.9921259842519685  
t-statistic: 1.1979862916737511  
p-value: 0.23317098509478104
```

Figure 25

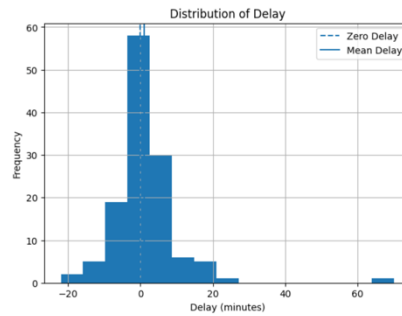


Figure 26

3.4. Machine Learning Models

To predict actual travel duration, several machine learning models were implemented and compared. Among these models, Linear Regression achieved the best overall performance with an R^2 score of 0.914 as shown in Figure 27, meaning that the model was able to explain approximately 91.4% of the variance in travel duration data. In addition, the model produced an average prediction error of nearly 5.46 minutes, indicating a high level of accuracy.

```
MAE: 5.46  
RMSE: 7.76  
R2: 0.914  
Meaning of R2: This model explains approximately 91.4% of the variation in actual travel duration.  
Comment: This indicates high predictive power.
```

Figure 27

Random Forest demonstrated a very similar performance with an R^2 score of 0.909. This suggests that the dataset contains some non-linear relationships; however, the improvement provided by a more complex ensemble model was not significantly higher than the simpler linear approach.

```
MAE: 5.36  
RMSE: 7.97  
R2: 0.909  
Meaning of R2: This model explains approximately 90.9% of the variation in actual travel duration.  
Comment: This indicates high predictive power.
```

Figure 28

K-Nearest Neighbors (KNN) also generated strong results with an R^2 value close to 0.90, further confirming that travel duration can be predicted successfully using the selected features.

On the other hand, the Decision Tree model showed the weakest performance with an R^2 score of 0.784. This indicates that a single-tree structure was not sufficient to fully capture the complexity and variability within the dataset.

Overall, Linear Regression and Random Forest emerged as the most effective models in this study. The findings demonstrate that travel duration can be predicted with high accuracy by using variables such as distance, traffic level, and estimated travel duration.

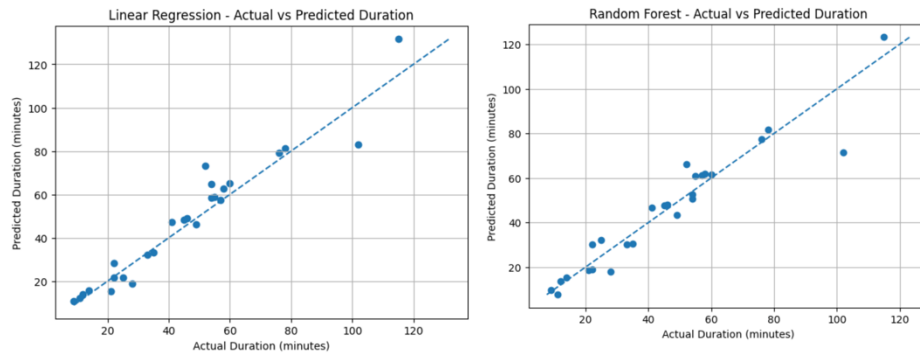


Figure 29

4. Findings

The analysis revealed several important insights regarding the factors affecting travel duration and the reliability of estimated travel times.

First, traffic conditions were identified as one of the most influential factors on actual travel duration. Both the exploratory data analysis and the ANOVA test demonstrated that increasing traffic levels significantly increase travel time. Trips under medium and high traffic conditions showed noticeably longer durations and greater variability compared to low traffic conditions. This indicates that traffic intensity not only increases travel duration but also reduces the predictability of trips.

The number of traffic lights showed only a weak relationship with travel duration, suggesting that traffic lights alone are not sufficient to explain overall trip variability.

Another important finding concerns the accuracy of Google Maps estimated travel durations. The correlation between estimated and actual durations was extremely high (approximately 0.97), indicating a very strong linear relationship. Furthermore, the paired t-test showed no statistically significant difference between estimated and actual travel times. Similarly, the one-sample t-test revealed that the average delay was not significantly different from zero. Together, these findings suggest that Google Maps predictions are generally reliable.

The analysis also showed that rush hour alone was not a statistically significant factor affecting travel duration. Although rush-hour trips sometimes exhibited longer durations, the independent t-test indicated that the differences between rush-hour and non-rush-hour trips were not statistically significant. This suggests that directly measured traffic level provides more meaningful information than simply categorizing trips based on time periods.

Machine learning models further demonstrated that travel duration can be predicted successfully using relatively simple features such as distance, traffic level, and estimated duration. Among the tested models, Linear Regression achieved the best overall performance with an R^2 score of 0.914, while Random Forest and KNN also produced highly accurate predictions. Decision Tree showed comparatively lower performance,

indicating that a single-tree structure was insufficient to fully capture the complexity of the dataset.

Overall, the findings of this project show that travel duration is strongly influenced by traffic conditions and travel distance, while estimated navigation times remain highly reliable. In addition, the machine learning results demonstrate that daily travel duration can be predicted with high accuracy using a limited number of easily obtainable variables.

5. Limitations and Future Work

One of the main limitations of this study is the relatively small dataset size. The analysis was based on 127 observations collected over a limited time period of approximately two weeks. While this amount of data was sufficient for exploratory analysis and basic machine learning modeling, a larger dataset collected over longer periods could provide more robust and generalizable results.

Another limitation is that the dataset was self-collected using trips made by the author, family members, and friends. As a result, the data may reflect specific driving habits, and traffic conditions, which may limit the generalizability of the findings to other cities or other drivers. In addition, some variables such as traffic level were categorized manually, which may introduce subjective bias into the analysis.

The project also focused on a limited set of features, including distance, traffic level, traffic lights, and estimated duration. However, travel duration may also be influenced by additional factors such as weather conditions, road quality, accidents, driver behavior, fuel consumption, or real-time traffic incidents. Including such variables could improve the explanatory and predictive performance of the models.

Although the machine learning models achieved high prediction accuracy, the models used in this study were relatively basic and focused primarily on regression-based approaches. More advanced machine learning or deep learning methods could potentially capture more complex patterns within the data.

For future work, the dataset could be expanded by collecting data over several months and across different locations to improve diversity and reliability. Real-time traffic APIs and weather data could also be integrated into the dataset to enrich the analysis.

Another possible extension would be developing a real-time travel duration prediction system or dashboard that dynamically predicts travel time based on current traffic and environmental conditions. This could transform the project from an analytical study into a practical transportation support tool.

Overall, despite its limitations, this project successfully demonstrates how data science techniques can be applied to real-world transportation problems and provides a strong foundation for more advanced future studies.